Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization Across Four Datasets and Required Properties for a Well-Formed Measure

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Abstract

Kinase inhibitor selectivity is quantified using several competing definitions – S-score, selectivity entropy, Gini coefficient, and ratio-based measures, which are applied interchangeably without justification. Across four profiling datasets spanning three assay technologies (68-704 compounds each), we show these definitions cluster into two families measuring different properties (cross-family r = 0.14-0.62), and that rank instability is predictable from binding profile shape. We trace instability to three sources: definitional noise below the detection threshold, near-tied top targets, and distribution-ratio disagreement for promiscuous compounds with a dominant target. We derive four properties that any well-formed selectivity measure should satisfy, show that no existing definition satisfies all four, and propose a measure that does and recovers its full-panel ranking from about half a panel.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with over 80 FDA-approved agents and several hundred in clinical development.[1] Because the human kinome is comprised of more than 500 proteins sharing a highly conserved ATP-binding site,[2] the design of inhibitors with controlled target profiles remains a central challenge. Selectivity is a primary criterion in lead optimization and a key determinant in the drug approval process.[3]

Despite its importance, no single consistent definition of selectivity exists. The field employs at least four distinct quantitative measures: the S-score,[4] selectivity entropy,[5] Gini coefficient,[6] and ratio-based measures including the CATDS score.[7] These definitions are applied interchangeably in the literature without systematic justification, and their parameter values (the activity threshold in the S-score, the baseline in the entropy and Gini coefficient, the reference off-target in ratio-based measures) are chosen by convention rather than principle. Standardizing them is a prerequisite for integrating selectivity data across laboratories and platforms.

Bosc et al.[8] showed that these definitions can produce substantially different compound rankings on the same dataset, and the KInhibition platform[9] explicitly acknowledged that conflicting results across metrics are common. However, the structural sources of this disagreement, such as "which binding profile properties determine when definitions agree or conflict, and why?", have not been characterized.

The absence of a principled basis for selectivity measurement is not merely a practical inconvenience. Go/no-go decisions worth hundreds of millions of dollars in development costs are made on the basis of selectivity assessments whose sensitivity to definitional choice is unknown. The lack of consistency across applications of selectivity definitions for different compounds makes comparison very challenging.

Our analyses suggest that existing selectivity definitions are, in some cases, measuring entirely different qualities of binding profiles, and therefore cannot be easily converted from one to another. In information theory, Shannon[10] settled which function should measure uncertainty by writing down a few properties a sensible measure must have (e.g, that it should be maximal when all outcomes are equally likely) and showing that only entropy satisfies them. In this work, we follow the first step of this precedent: we characterize how the competing measures actually behave, then state the properties a good measure should have.

We use four large profiling datasets for kinase inhibitor selectivity that span three assay technologies in our characterization. In service of the overarching goal of developing a principled, standardized formula for defining selectivity, our contributions are as follows:

(i) we show that existing definitions cluster into two families: distribution-based (S-score, entropy, Gini) and ratio-based (CATDS and variants), which correlate weakly with one another across all four datasets (cross-family $r = 0.14$–0.62) and measure distinct properties of binding profiles;

(ii) we identify three sources of instability: definitional noise for compounds below the assay detection threshold, ratio-specific instability driven by near-tied top targets, and distribution-ratio disagreement for compounds with one dominant target and broad moderate off-target activity;

(iii) we propose four required properties: (D1) a reliability threshold condition, (D2) bounded gap sensitivity, (D3) distributional consistency, (D4) and monotonicity under weak off-target addition. We then show that no existing definition satisfies all four simultaneously;

(iv) we propose a candidate measure: a gated, smooth-baseline effective-target-number that satisfies all four properties and recovers its full-panel ranking from roughly half a typical panel.

Related Work

Existing Selectivity Metrics and Their Comparison

Several metrics have been proposed to quantify kinase inhibitor selectivity from profiling panel data. Karaman et al.[4] introduced the selectivity score (S-score), defined as the fraction of panel kinases inhibited above a fixed concentration threshold. Uitdehaag and Zaman[5] proposed the selectivity entropy, which treats the normalized binding profile as a probability distribution and computes its Shannon entropy. Low entropy indicates activity concentrated on few targets. Graczyk[6] adapted the Gini coefficient from economics, treating the distribution of a compound's activity across the panel as an inequality distribution. A value near 1 indicates activity concentrated on few kinases and near 0 indicates activity spread evenly. Klaeger et al.[7] introduced the concentration- and target-dependent selectivity (CATDS) score, which quantifies the reduction in binding to a given target relative to the summed reduction across all detected targets at a specified inhibitor concentration. Bosc et al.[8] subsequently proposed two further metrics: the window score and ranking score. They performed a comparative analysis of all five definitions on the Davis and Anastassiadis datasets, showing that compound rankings differ substantially depending on the metric applied and that certain compounds cannot be scored under some definitions when threshold conditions are not satisfied. For example, ratio-based scores are undefined for a compound with no kinase above the activity threshold, since there is then no primary target or off-target to compare.

Reconciliation attempts have remained limited in scope. Bajorath et al.[11] compared CATDS-based selectivity assessments from Klaeger et al. with activity data mined from ChEMBL for the same compounds. They found broad consistency between the two sources but noted that no simple relationship between selectivity and clinical performance was apparent. The KInhibition platform[9] explicitly acknowledged that multiple metrics yield conflicting results for the same compound but addressed this through a unified query interface rather than theoretical analysis.

A common limitation of these studies is that each metric is evaluated at fixed parameter values, so the sensitivity of rankings to parameter choice has not been systematically characterized. The structural properties of binding profiles that determine when definitions agree or disagree have not been identified. The present work addresses both gaps.

Axiomatic Approaches to Ranking and Measurement

Axiomatic characterization of measurement functions is well-established in adjacent fields. Shannon[10] demonstrated that the entropy of a probability distribution is uniquely determined, up to a multiplicative constant, by four requirements: continuity, maximality on the uniform distribution, expansibility, and additivity over independent subsystems. Khinchin[12] subsequently strengthened this uniqueness result. The selectivity entropy of Uitdehaag and Zaman applies the Shannon formula to normalized binding profiles but provides no analogous axiomatic justification. The axioms that single out Shannon entropy, such as maximality on the uniform distribution and additivity over independent subsystems, characterize a measure of uncertainty rather than a measure of selectivity. It is therefore not known whether the selectivity entropy is the unique function satisfying reasonable selectivity-specific properties, or one of many equivalent alternatives.

Arrow[13] established that no social welfare function satisfies simultaneously the requirements of unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship. The theorem is relevant here in two respects: it demonstrates that axiomatic analysis of a ranking concept can yield impossibility rather than uniqueness, and it shows that competing ranking functions reflect genuine tradeoffs between properties that cannot all be achieved simultaneously. Singleton and Booth[14] applied this methodology to truth discovery, i.e. the problem of identifying true facts from conflicting reports of unknown reliability, posing it as a social choice problem, formulating candidate axioms, and proving that a subset cannot be jointly satisfied. The structure of that problem is closely analogous to selectivity measurement, where multiple definitions produce conflicting rankings and the question of which to trust is unresolved.

To our knowledge, this axiomatic style of analysis has not previously been applied to any selectivity or promiscuity measure in drug discovery. The present work proposes empirically motivated required properties as a foundation for such an analysis.

Methods

Datasets

Two publicly available kinase inhibitor profiling datasets were used. The Davis dataset[15] comprises binding affinities (K_d) for 68 small-molecule kinase inhibitors measured against 433 human kinase domains. Affinity values are reported as dissociation constants and were converted to p$K_d = -\log_{10}(K_d/10^9)$ prior to analysis, following the convention of the DeepDTA benchmark.[16] The resulting affinity matrix is complete (68×433, 29,444 entries). with all non-binding pairs assigned the detection-floor value p$K_d = 5.0$ (corresponding to a K_d of 1 mM). This is the standard convention for the dataset[8,16] and slightly underestimates true non-binding affinity.

The Klaeger dataset[7] comprises binding affinities for 243 clinical kinase inhibitors profiled against the human kinome using Kinobeads chemical proteomics. Apparent dissociation constants (K_d^{app}) were derived from dose–response curves across eight inhibitor concentrations in cancer cell lysates. Only high-confidence measurements (as annotated by the original authors) were retained, yielding 5,123 drug–target pairs for 222 drugs and 343 kinase targets. Affinity values were converted to pK_d using the same formula as above. Missing entries, representing kinase–inhibitor combinations not detected in the assay, were assigned the background value p$K_d = 5.0$. The resulting Klaeger matrix is 222×343. The fraction of active entries (pKd > 6, i.e. $K_d < 1$ μM) is 17.9% for Davis and 3.6% for Klaeger, and are recorded at the background value whenever weak interactions fall below the limit of detection.

These reflect the more stringent detection threshold of the chemoproteomic assay.

To test whether our findings depend on the assay technology, we added two further datasets that measure kinase inhibition by different means. The Anastassiadis dataset[17] profiles 178 inhibitors against 300 kinases in a single-concentration (0.5 μM) radiometric activity assay, reported as percent residual kinase activity relative to a solvent control. We converted these to percent inhibition (100 minus residual activity, clipped to $[0, 100]$, where higher values denote stronger inhibition), with the 1.1% missing entries set to zero inhibition. The Metz dataset[18] reports pK_i affinities for a large medicinal-chemistry campaign using the same negative-log scale as pK_d. We restricted it to the 704 compounds profiled against at least 50 of the 172 kinases, assigning the assay floor p$K_i = 4.0$ to unmeasured combinations. The four datasets thus span three assay technologies (competition binding, chemoproteomics, and functional activity) and two readout types (affinity and single-concentration inhibition). Each selectivity definition is computed on the native readout of its dataset and our analyses concern rank correlations between definitions and rank instability, which are both invariant to a common monotone rescaling within a dataset. For this reason, the definitions remain directly comparable across datasets. For the percent-inhibition and pK_i datasets, the S-score threshold and entropy/Gini baseline grids were shifted to span each dataset's active range, while the ratio off-target index $k \in \{1, \ldots, 5\}$ was left unchanged (it indexes ranked off-targets and so needs no rescaling).

We excluded two additional candidate datasets. The Gao screen[19] is a single-concentration functional inhibition assay of the same readout type as the Anastassiadis dataset already included, and its raw data are not openly redistributable. Including it would add a second instance of an assay type rather than a new modality. The Patricelli study[20] profiles only a small number of inhibitors against native kinases as a proof of concept for the KiNativ platform. A ranking-stability analysis requires many compounds to produce a meaningful ranking, so this dataset is not suited to the present comparison.

Selectivity Definitions and Parameterization

We implemented four selectivity definitions, each including parameters whose values are not standardized in the literature. We discuss each definition below. Throughout, a compound's binding profile is the vector of its affinities $\mathrm{p}K_{d,i}$ across the N panel kinases, with higher $\mathrm{p}K_d$ denoting tighter binding.

S-score. Karaman et al.[4] defined the selectivity score as the fraction of the panel that a compound inhibits: the number of kinases "bound" divided by the number tested, where a kinase counts as bound when the compound reduces its activity below a fixed cutoff at a single screening concentration (the original work reports $S(35)$, $S(10)$ and $S(1)$ for cutoffs of 35%, 10% and 1% of control activity). We implemented the same fraction, expressing the bound criterion as an affinity threshold τ on $\mathrm{p}K_d$:

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mathrm{p}K_{d,i} > \tau]. \tag{1}$$

Karaman et al. defined the cutoff on percent of control activity (35%, 10%, 1%). Because we work on the $\mathrm{p}K_d$ scale, the equivalent "bound" criterion is an affinity threshold τ, and sweeping it reproduces their family of cutoffs. We used $\tau \in \{5.5, 5.75, \ldots, 8.0\}$ (11 values), spanning K_d cutoffs from approximately 300 nM to 10 nM.

Selectivity entropy. Uitdehaag and Zaman[5] treat the compound's binding across the panel as a probability distribution and report its Shannon entropy. A profile concentrated on few kinases has low entropy and is selective. In the original definition the weight on each kinase is its association constant $K_{a,i} = 1/K_{d,i}$ (the share of compound that would be bound to kinase i in an equimolar mixture), so $p_i = K_{a,i}/\sum_j K_{a,j}$. We implemented the entropy on the same log-affinity scale used by the other metrics, weighting each kinase by its affinity in excess of a baseline β:

$$H(\beta) = -\sum_{i=1}^{N} p_i \log_2 p_i, \quad p_i = \frac{\max(\mathrm{p}K_{d,i} - \beta,\, 0)}{\sum_j \max(\mathrm{p}K_{d,j} - \beta,\, 0)}, \tag{2}$$

where β is the affinity below which binding is treated as negligible. This makes β an explicit parameter (its under-specification motivates property D3 below). We swept $\beta \in \{5.0, 5.25, \ldots, 6.5\}$ (7 values). The association-constant and log-affinity weightings are distinct reasonable choices that can rank compounds differently, which is itself an instance of the definitional instability this paper characterizes. We quantify the difference below.

Gini coefficient. Graczyk[6] adapted the Gini coefficient from economics: kinases are sorted by activity and the cumulative fraction of total binding is plotted against the cumulative fraction of kinases (a Lorenz curve). The Gini coefficient, which measures how unequally is binding distributed across the kinome, is twice the area between this curve and the diagonal of equal distribution. The original work computes it from percent-inhibition values. We applied the standard discrete Gini formula to the same baseline-subtracted log-affinities used for the entropy:

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot x_{(i)}}{N\sum_{i=1}^{N} x_{(i)}} - \frac{N+1}{N}, \qquad x_{(i)} = \max(\mathrm{p}K_{d,i} - \beta,\, 0) \text{ sorted ascending.} \tag{3}$$

We swept β over the same range as for the entropy. A compound whose binding is concentrated on one kinase has a highly unequal distribution and G near 1, while a compound that binds many kinases equally has G near 0. Higher values therefore indicate greater selectivity.

Ratio. Classical pharmacological selectivity is pairwise: how much more tightly does a compound bind its primary target than an off-target? We implemented this as the fold-selectivity, in log units, between the primary target and the k-th ranked off-target:

$$R(k) = \mathrm{p}K_{d,(1)} - \mathrm{p}K_{d,(k+1)} = \log_{10} \frac{K_{d,(k+1)}}{K_{d,(1)}}, \tag{4}$$

where $\mathrm{p}K_{d,(j)}$ is the j-th largest affinity in the profile and $k \in \{1, 2, 3, 4, 5\}$ selects the reference off-target. This is the prototypical member of the ratio family, which also includes the concentration- and target-dependent selectivity (CATDS) score of Klaeger et al.[7] Members of the ratio family of selectivity definitions are concentration-dependent fractions quantifying the binding reduction at one target relative to the summed reduction across all targets. We use the simpler fold-selectivity because the reference off-target index k is then an explicit parameter whose effect can be studied directly. A larger affinity gap to the nearest competitor means greater selectivity, so higher $R(k)$ indicates greater selectivity.

Faithfulness to the original definitions. The S-score and the ratio are computed in the form given by their original sources (a panel-hit fraction and a primary-to-off-target affinity ratio, respectively). For the entropy and the Gini coefficient, we used the baseline-subtracted log-affinity profile in place of the original association constants (entropy) and percent-inhibition values (Gini), so that all four metrics operate on one common scale, and so that the baseline β appears as an explicit parameter. To confirm that this choice does not silently determine our conclusions, we recomputed the entropy and Gini in forms closer to the originals (association-constant weighting) and compared the resulting rankings with the log-affinity forms. The two operationalizations differ substantially — for entropy, Spearman $r = 0.42$ (Davis) and -0.32 (Klaeger), while for Gini, $r = 0.11$ and -0.46 (Supporting Information, Figure S1). This confirms that the operationalization of a named metric is itself a source of the definitional instability we report, not a neutral implementation detail.

Rank Stability Analysis

For each definition and each parameter value, compounds were ranked from most selective (rank 1) to least selective (rank n, where n is the number of compounds in the dataset). Ties were broken by average rank. This yielded 30 rank vectors per dataset: 11 for the S-score, 7 for entropy, 7 for Gini, and 5 for ratio.

For each compound, rank instability was quantified as the standard deviation of its rank across all 30 configurations (overall instability) and separately across parameter configurations within each definition family (within-family instability). Cross-definition agreement was assessed using Spearman rank correlation between the median rank vectors of each definition family. The median was taken across parameter values within each family prior to correlation to obtain a single representative ranking per definition.

Associations between rank instability and binding profile features were assessed using Spearman correlation. Profile features computed for each compound included: the number of active kinases (p$K_d > 6$), the gap between the top and second-ranked affinities, the standard deviation and range of active affinities, and the absolute affinity of the second-ranked target. All correlations are reported with two-tailed p-values. Significance thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$ are denoted *, **, and *** respectively. Analyses were performed in Python 3.10 using NumPy, pandas, and SciPy.[21–23] A secondary analysis relating selectivity rankings to clinical adverse-event outcomes (FAERS report rates and FDA-label discontinuation rates for the FDA-approved Klaeger compounds) is summarized in the Limitations section and reported in full in the Supporting Information section.

Results

Selectivity Definitions Cluster into Two Families

Across 30 parameter configurations (11 S-score thresholds, 7 entropy baselines, 7 Gini baselines, 5 ratio off-target indices), compound rankings were computed for all drugs in both datasets. Spearman correlations between the median ranking of each definition family reveal a consistent two-cluster structure (Table 1). On the Davis dataset, S-score, entropy, and Gini are mutually correlated ($r = 0.91$–0.999), while ratio correlates substantially less with each ($r = 0.34$–0.45). On the larger Klaeger dataset, the same structure is preserved but with lower within-cluster correlations ($r = 0.75$–0.91), reflecting greater diversity of binding profiles among the 222 clinical compounds. The ratio definition is the consistent outlier in both datasets, with cross-family correlations of $r = 0.27$–0.62.

Table 1: Spearman correlations between median rankings by definition family. Upper triangle: Davis ($n = 68$). Lower triangle: Klaeger ($n = 222$).

	S-score	Entropy	Gini	Ratio
S-score	—	0.949	0.954	0.454
Entropy	0.910	—	0.999	0.343
Gini	0.745	0.869	—	0.343
Ratio	0.265	0.480	0.617	—

This two-family structure reflects a fundamental difference in what the definitions measure. The S-score, entropy, and Gini all quantify the concentration of activity across the full kinase panel. They differ in functional form but respond to the same underlying property of the binding profile. The ratio definition, by contrast, quantifies the gap between the primary target and a specific off-target, ignoring the remainder of the binding profile entirely. A compound that strongly inhibits one kinase and moderately inhibits twenty others will be ranked as highly selective by ratio (large gap to the second target) but as moderately promiscuous by entropy and Gini (activity distributed across many targets).

The Two-Family Structure Replicates Across Assay Technologies

The Davis and Klaeger pattern of Table 1 extends to the two further datasets, across three assay technologies and both readout types (Figure 1). In all four, the S-score, entropy, and Gini remain mutually correlated (within-family Spearman $r = 0.74$–0.99) while the ratio definition is the consistent outlier. Its correlation with the distribution-based family is 0.52–0.56 on the Anastassiadis functional-activity dataset and only 0.14–0.19 on the Metz pK_i dataset. The degree of separation varies with the assay, but in every case the ratio definition measures something the distribution-based definitions do not.

The instability findings also replicate. Compounds with no active kinases (Type 1, below) are markedly more rank-unstable than active compounds wherever such compounds exist: mean rank $\sigma = 37.6$ vs. 22.0 on Anastassiadis and 178.7 vs. 108.2 on Metz, matching the 73.8 vs. 32.4 contrast on Klaeger (Davis contains no zero-active compounds). Within the distribution-based family, the number of active kinases negatively predicts entropy instability in every dataset ($r = -0.28$ to -0.46), and the top$_1$–top$_2$ affinity gap predicts ratio instability on Davis, Klaeger, and Metz ($r = -0.45$, -0.34, -0.26; all $p < 0.001$), as detailed below.

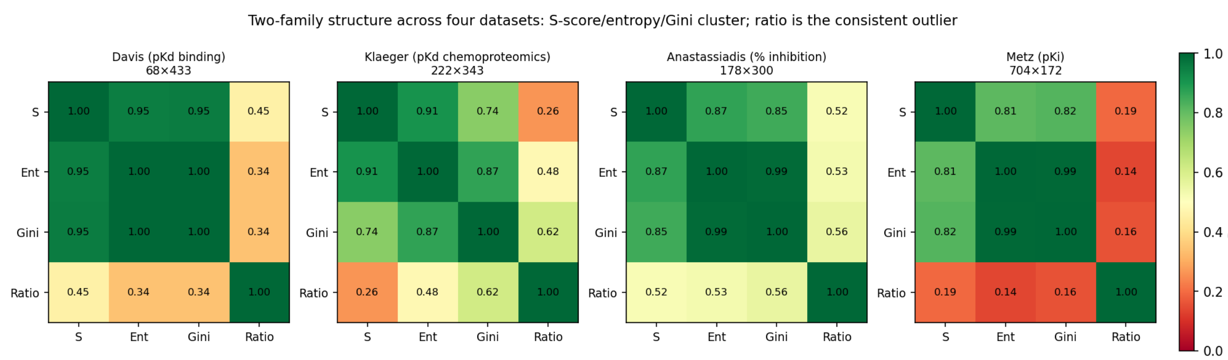

Figure 1: Spearman correlations between the median rankings of the four selectivity definitions, computed independently on each of the four datasets (compound × kinase panel sizes shown). In every dataset the S-score (S), entropy (Ent), and Gini cluster together while the ratio is the consistent outlier (the ratio row and column are the least green throughout), confirming the two-family structure across competition-binding, chemoproteomic, and functional-activity assays.

Rank Instability Is Structured and Predictable

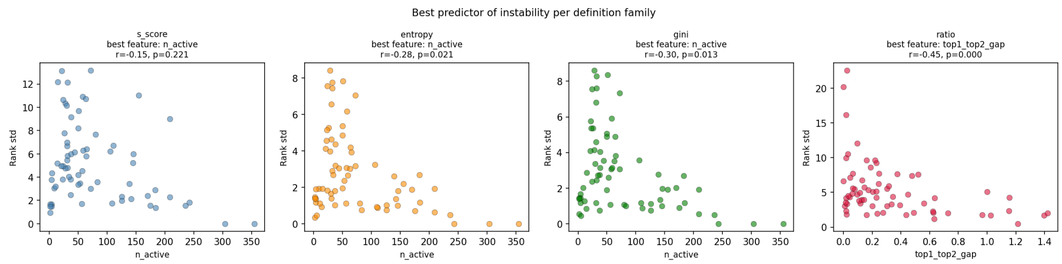

Figure 2: Best predictor of rank instability within each definition family (Klaeger dataset, active compounds, $n = 206$). Ratio instability is best predicted by the top_1–top_2 affinity gap; entropy, Gini, and S-score instability are best predicted by features of the active affinity distribution.

Overall rank instability (measured as the standard deviation of a compound's rank across all 30 configurations) varies substantially across compounds (Klaeger: mean $\sigma = 35.4$, range 7.1–91.6; Davis: mean $\sigma = 8.5$, range 1.6–19.3). Instability is not uniformly distributed: within each definition family, specific binding profile features predict which compounds receive unstable rankings (Table 2).

Within the ratio family, the gap between the top and second-ranked affinities (top_1–top_2 gap) is the strongest predictor of instability (Figure 2) ($r = -0.446$, $p < 0.001$ on Davis, and $r = -0.341$, $p < 0.001$ on Klaeger). Compounds with near-identical affinities for their two best-bound kinases receive highly variable ratio rankings because the choice of reference off-target ($k = 1$ vs. $k = 2, \ldots, 5$) determines whether the compound appears selective or promiscuous.

Within the entropy and Gini families, the number of active kinases ($\text{p}K_d > 6$) negatively predicts instability ($r = -0.279$, $p < 0.05$ for entropy on Davis, $r = -0.455$, $p < 0.001$ on Klaeger). Compounds with many active kinases have well-defined broad profiles whose entropy and Gini scores are stable across baseline choices, while compounds with few active kinases are sensitive to the baseline because small changes can move targets in or out of the active set, altering the distribution substantially. The active affinity range is also a significant predictor for the S-score and entropy on Klaeger ($r = -0.46$ and -0.32, both $p < 0.001$, Gini

$r = -0.16$, $p < 0.05$). It is consistent with the interpretation that heterogeneous profiles (whose affinities span a wide range) yield more reproducible rankings across parameter choices than flat profiles (in which many near-equal affinities reshuffle as the parameter changes).

S-score instability is not significantly predicted by any profile feature on either dataset, suggesting that it is driven primarily by where compounds fall relative to the threshold rather than by intrinsic profile shape.

Table 2: Spearman correlations between binding profile features and within-family rank instability on the Klaeger dataset ($n = 206$ active drugs). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Feature	S-score	Entropy	Gini	Ratio
n_{active}	-0.350^{***}	-0.455^{***}	-0.166^{*}	-0.040
top_1–top_2 gap	$+0.058$	$+0.128$	$+0.109$	-0.341^{***}
Active affinity std	-0.386^{***}	-0.215^{**}	-0.176^{*}	-0.107
Active affinity range	-0.458^{***}	-0.318^{***}	-0.163^{*}	-0.116
top_2 pK_d	-0.394^{***}	-0.282^{***}	-0.268^{***}	$+0.186^{**}$

Three Mechanistically Distinct Sources of Instability

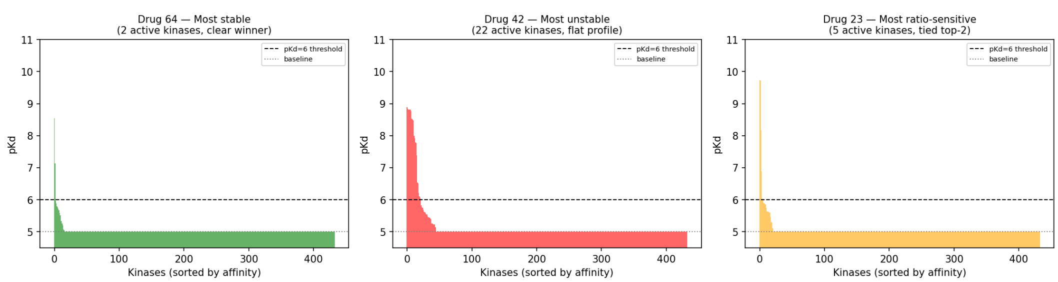

Figure 3: Representative binding profiles illustrating the three instability types. Left: Drug 64 (Davis), most stable ($\sigma = 1.6$), two active kinases. Center: Drug 42 (Davis), most unstable ($\sigma = 19.3$), 22 active kinases at near-identical affinities. Right: Drug 23 (Davis), most ratio-sensitive, five active kinases including two near-tied top targets (Δp$K_d = 0.019$). Dashed line: p$K_d = 6$ activity threshold; dotted line: p$K_d = 5$ assay floor.

Examination of the most and least unstable compounds in the Klaeger dataset reveals three structurally distinct sources of rank instability (Figure 3), each implicating a different aspect of how the definitions are constructed.

Type 1 — Zero-active compounds. The five most unstable Klaeger compounds (e.g. AZD-8055, BMS-911543, Filgotinib, rank $\sigma > 80$) all have zero active kinases at the p$K_d > 6$ threshold. These compounds have no meaningful binding to any profiled kinase at sub-micromolar affinity, yet all four definitions still assign them numerical selectivity scores based on the detection floor values. The resulting rankings are arbitrary: different parameterizations produce different orderings of compounds that are, in practice, indistinguishable from one another. Mean rank instability for zero-active compounds is $\sigma \approx 74$, compared with $\sigma \approx 32$ for compounds with at least one active kinase ($p < 0.001$, Mann–Whitney U test). This pattern is observed only in the Klaeger dataset: all Davis compounds have at least one active kinase, consistent with Davis's fixed-concentration screening design, which reports a K_d only when activity is first detected at 10 μM.

Type 2 — Near-tied top targets. Among active compounds, ratio-specific instability is strongly predicted by a small top$_1$–top$_2$ gap ($r = -0.341$, $p < 0.001$). When a compound's two best-bound kinases have near-identical affinities, the ratio is highly sensitive to which is designated the primary target and which the reference off-target, because a small parameter change can swap their roles. Drug 23 in the Davis dataset (top$_1$–top$_2$ gap = 0.019 pKd units) is the extreme case: its two best-bound kinases are essentially indistinguishable, so it receives maximally unstable ratio rankings.

Type 3 — Broad profiles with a dominant primary target. The largest entropy-ratio disagreements in the Klaeger dataset arise for compounds with high n_{active} and a large top$_1$–top$_2$ gap (e.g. Dovitinib: $n_{\text{active}} = 26$, gap = 1.935; BMS-690514: $n_{\text{active}} = 20$, gap = 1.985). These compounds have one clearly dominant target but also bind many kinases at moderate affinity. The ratio definition scores them as highly selective (large gap between primary and secondary target), while the entropy and Gini definitions score them as moderately promiscuous (activity spread across many targets). Neither characterization is incorrect, as they reflect genuinely different questions about selectivity.

Conversely, the most stable compounds in both datasets (e.g. XL-228, $\sigma = 7.1$; PF-

03814735, $\sigma = 8.1$) are highly promiscuous, with 40–57 active kinases. When a compound binds nearly the entire profiled kinome, all definitions rank it as non-selective regardless of parameterization, producing low instability.

Panel Size Dependence of Selectivity Rankings

To assess the stability of selectivity rankings as a function of kinase panel size, we performed a subsampling analysis on the Klaeger dataset. The Davis dataset was not used for this analysis for two reasons: its complete matrix design (every compound tested against every kinase by construction) does not reflect the sparse coverage of real profiling panels, and subsampling kinases from Davis changes the apparent number of active compounds in ways that confound the panel size effect with the zero-active instability characterized in the instability taxonomy above. The Klaeger dataset, derived from chemoproteomic dose–response profiling with genuine sparse kinase coverage, is the more appropriate substrate for this question. For each panel size $p \in \{50, 80, \ldots, 320, 343\}$, we drew 50 random subsets of p kinases from the full panel of 343, computed each selectivity definition on the subpanel, and measured agreement with the full-panel ranking using Spearman correlation.

The four definitions differ substantially in their panel size requirements (Figure 4). Selectivity entropy reaches mean $r = 0.90$ against the full-panel ranking at approximately 110 kinases, and S-score at approximately 140 kinases. The Gini coefficient requires approximately 290 kinases to reach the same threshold. The ratio definition does not reach $r = 0.90$ until the panel contains 320 or more kinases, and achieves $r = 0.888$ even at 320 kinases — only 23 below the full panel size.

This result has a direct practical implication. Standard commercial kinase selectivity panels typically profile 50–100 kinases.[8] At these panel sizes, entropy- and S-score-based rankings are reasonably stable ($r \approx 0.76$–0.85), but Gini and ratio rankings are effectively uncorrelated with what would be obtained on a full kinome panel ($r < 0.14$ for Gini, $r < 0.00$ for ratio at $p = 50$). Ratio rankings in particular are unreliable at any panel size commonly used in practice, because the identity of the reference off-target (k-th ranked kinase) changes substantially as panel composition changes. This is further evidence that ratio-based definitions measure a qualitatively different property from distribution-based definitions, and reinforces D2 (bounded gap sensitivity) as a necessary property.

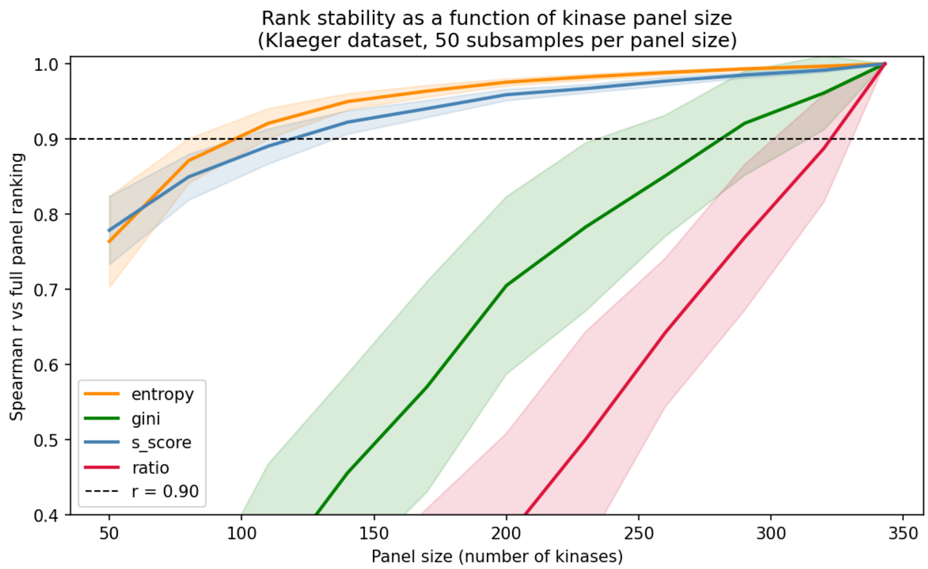

Figure 4: Mean Spearman correlation between rankings computed on random kinase subpanels and rankings on the full Klaeger panel (343 kinases), as a function of subpanel size. Shaded bands indicate ± 1 standard deviation across 50 subsamples. The dashed line marks $r = 0.90$. Entropy and S-score rankings stabilize at approximately 110–140 kinases; Gini requires approximately 290 kinases; ratio does not reach $r = 0.90$ at any practically used panel size.

Required Properties for a Well-Formed Selectivity Measure

The empirical findings of the Results section motivate four properties that any well-formed kinase inhibitor selectivity measure should satisfy. Each property is stated below together with the observed failure mode that motivates it.

Let $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{R}^N$ denote a compound's binding profile across N kinases, where $x_i = \mathrm{p}K_{d,i}$. A selectivity measure is a function $f : \mathbb{R}^N \to \mathbb{R}$, where higher values indicate greater selectivity. A selectivity ranking over a set of compounds is induced by ordering their f-scores.

D1 — Reliability threshold. There exists a threshold $\tau^* > 0$ such that $f(\mathbf{x})$ is undefined, or explicitly flagged as unreliable, when $\max_i x_i \leq \tau^*$.

Motivation. Compounds with no detected binding above the assay detection floor receive arbitrary selectivity scores under all four definitions studied here, producing rank instability with mean $\sigma \approx 74$ compared with $\sigma \approx 32$ for active compounds (Type 1 instability, above). Selectivity is a property of binding, so a compound that does not bind cannot be meaningfully ranked relative to compounds that do. No existing definition incorporates an explicit reliability condition: both the S-score and selectivity entropy assign scores to compounds with no sub-micromolar binding, treating detection floor values as informative.

D2 — Bounded gap sensitivity. For any compound profile $\mathbf{x}$ satisfying D1, a small perturbation to the identity of the reference off-target should produce a bounded change in $f(\mathbf{x})$. Formally, if f depends on the k-th ranked affinity $x_{(k)}$, then $|f(\mathbf{x}; k) - f(\mathbf{x}; k+1)|$ should be bounded by a function of $x_{(k)} - x_{(k+1)}$.

Motivation. Ratio-based definitions violate this property when $x_{(1)} \approx x_{(2)}$: a compound with near-identical affinities for its top two kinases receives a large ratio score when $k = 1$ (the gap to the next off-target may be large) but a near-zero score when $k = 2$ (the gap between the tied targets is negligible). This produces the Type 2 instability identified in the instability taxonomy above, where top_1–top_2 gap is the strongest predictor of ratio instability ($r = -0.341$, $p < 0.001$). Distribution-based definitions (entropy, Gini) satisfy a weaker version of this property because they are symmetric functions of the full set of normalized activities, with no term that singles out a particular ranked off-target.

D3 — Distributional consistency. The ranking induced by f should be stable under small perturbations to the activity baseline β used to define the active binding distribution.

Formally, for any two compounds $\mathbf{x}, \mathbf{y}$ and any $\epsilon > 0$, there exists $\delta > 0$ such that if $|\beta_1 - \beta_2| < \delta$ then the ranking of $\mathbf{x}$ relative to $\mathbf{y}$ under $f(\cdot; \beta_1)$ agrees with that under $f(\cdot; \beta_2)$.

Motivation. The entropy and Gini definitions are parameterized by a baseline β that determines which affinities contribute to the binding distribution. Compounds with few active kinases are sensitive to this choice because small changes in β can move individual kinases in or out of the active set, substantially altering the distribution. The negative correlation between n_{active} and entropy/Gini instability ($r = -0.47$ on Klaeger, $p < 0.001$) reflects this sensitivity. D3 requires that the ranking be robust to the choice of β within a reasonable neighborhood, which existing implementations do not guarantee.

D4 — Monotonicity under weak off-target addition. If compound $\mathbf{x}$ has a weak additional binding interaction added — formally, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ where $x_{N+1} \leq \tau^*$ — then $f(\mathbf{x}') \leq f(\mathbf{x})$. Adding a sub-threshold binding interaction should not increase a compound's selectivity score.

Motivation. The S-score and the Gini coefficient violate this property. Adding a kinase that the compound does not meaningfully bind increases the panel size without adding activity: this lowers the S-score's fraction of hit kinases and raises the Gini coefficient's inequality, in both cases inflating the apparent selectivity. A compound can therefore be made to look more selective simply by profiling it against more inactive kinases — a serious problem when panels differ in size or composition across experiments, as between Davis and Klaeger. The entropy and the ratio satisfy D4: a sub-threshold target adds negligible weight to the entropy's normalized distribution and falls at or below the ratio's reference off-target, so neither score increases (verified empirically in the next section).

Table 3 summarizes which definitions satisfy each property. No existing definition satisfies all four simultaneously, but the candidate measure introduced next does.

Table 3: Satisfaction of the four required properties by existing selectivity definitions and by the candidate measure of Section . ✓: satisfied (by construction, or verified empirically for the candidate). $\sim$: partially satisfied or satisfied under additional assumptions. $\times$: violated in general.

Property	S-score	Entropy	Gini	Ratio	Candidate
D1 (reliability threshold)	×	×	×	×	✓
D2 (bounded gap sensitivity)	∼	✓	✓	×	✓
D3 (distributional consistency)	∼	×	×	✓	✓
D4 (monotonicity, weak off-target)	×	✓	×	✓	✓

A Candidate Measure

The four properties can indeed be satisfied, and the selectivity entropy is the natural starting point. Of the existing definitions it is the only one that already satisfies both D2 and D4 (Table 3): because it scores the whole normalized profile rather than a single pairwise gap, it neither privileges a reference off-target (D2) nor rewards the addition of non-binders (D4). It fails only D1, having no reliability gate, and D3, because its hard active/inactive cutoff $\max(x_i - \beta, 0)$ makes the ranking sensitive to the baseline. The ratio cannot be repaired this way, as it fails D2 by construction, and the S-score and Gini violate D4 intrinsically. The entropy is therefore the one existing measure that can reach all four properties through purely local changes. The candidate makes exactly two: it adds the D1 reliability gate, and it replaces the hard cutoff with a smooth (softplus) hinge that removes the baseline sensitivity. For a profile $\mathbf{x}$ with $\max_i x_i > \tau^*$ (and undefined otherwise, satisfying D1), define

$$w_i = T \log\bigl(1 + e^{(x_i - x_0)/T}\bigr), \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

where x_0 is the fixed assay detection floor and T a smoothing width in pK_d units. The score is minus the Shannon entropy of the smoothed, normalized profile — equivalently, minus the logarithm of the effective number of targets the compound binds.

This measure satisfies all four properties. D1 holds by the explicit reliability gate. D2 holds because f is a symmetric function of the full normalized profile and references no particular off-target. D3 holds because the smooth hinge has no hard active/inactive boundary, so no kinase abruptly enters or leaves the active set as x_0 shifts: empirically, the candidate's ranking is essentially invariant to the floor over $x_0 \in [4.5, 6.5]$ (worst-case pairwise Spearman $\rho = 0.99$), whereas the existing entropy's ranking can reverse over the same range ($\rho = -0.95$). D4 holds because the softplus weight vanishes below the floor and is monotone, so adding a sub-threshold target never raises the score (verified for 100% of Klaeger compounds).

A usable measure must also be recoverable from realistic panels. Applying the subsampling analysis of Section to the candidate (same grid and 50 random subpanels per size), the smallest panel at which its ranking reaches mean Spearman $\rho > 0.90$ with its full-panel ranking is $p^* \approx 170$ kinases (Figure 5), placing it in the fast regime alongside the entropy ($p^* \approx 110$) and S-score (≈ 140) and well ahead of the Gini (≈ 290) and the ratio (which does not reach $\rho = 0.90$ even at the full panel). The candidate thus recovers its full-panel ranking from roughly half a typical large panel.

We present this as a candidate, not a unique solution. The properties are necessary conditions, and we have not shown that they single out a measure of this form. The smoothing width T is itself a design choice ($T \approx 1$ gives the values above). Establishing whether the properties are jointly satisfiable only by measures of this family, and whether a uniqueness result analogous to Shannon's holds, is the subject of future work.

Discussion

The central finding of this work is that existing kinase inhibitor selectivity definitions do not measure the same quantity. The ratio definition and the distribution-based family (S-score, entropy, Gini) produce substantially different compound rankings (cross-family $r = 0.14$–0.62 across four datasets), and the sources of disagreement are mechanistically interpretable rather than random. This has practical consequences: a compound that ratio-based analysis designates as selective may be designated as moderately promiscuous by entropy-based

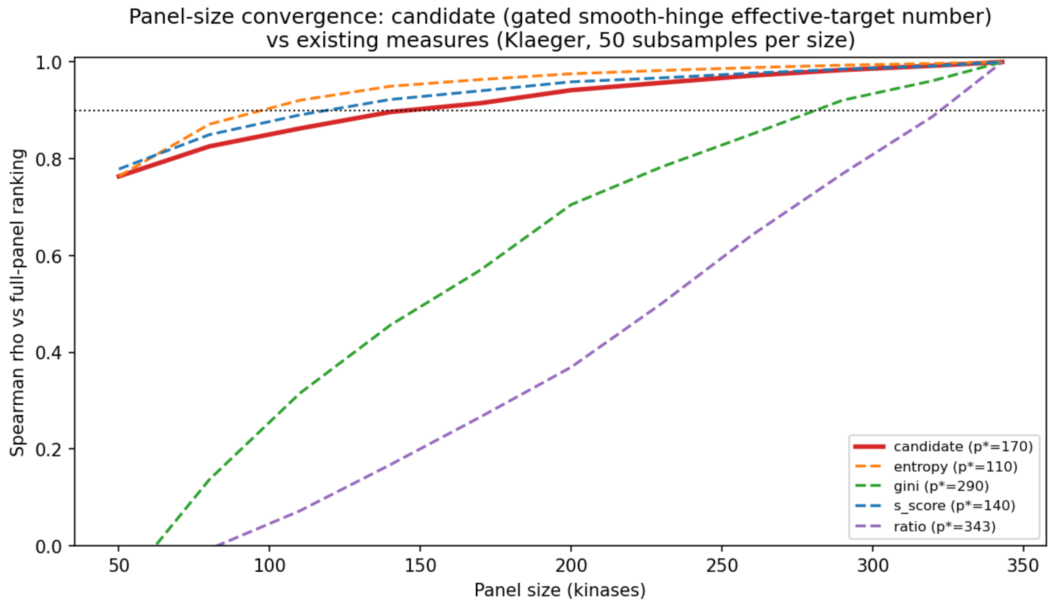

Figure 5: Panel-size convergence of the candidate measure compared with the four existing definitions (Klaeger, 50 random subpanels per size). Curves show the mean Spearman correlation between the ranking computed on a random subpanel and the full-panel ranking; p^* (legend) is the smallest panel reaching mean $\rho > 0.90$. The candidate converges in the fast regime ($p^* \approx 170$), comparable to the entropy and S-score and well ahead of the Gini and ratio.

analysis, and vice versa, depending on the shape of its binding profile. Go/no-go decisions in early drug discovery are routinely made on the basis of selectivity assessments without acknowledgment that the choice of definition materially affects the outcome.

The clustering of definitions into two families is not a statistical artifact but reflects a genuine difference in what is being measured. Distribution-based definitions ask: how concentrated is the compound's binding activity across the full profiled kinome? The ratio definition asks: how much more potently does the compound bind its primary target than a specified off-target? These are distinct questions with distinct answers for a large class of compounds — specifically, those with one clearly dominant target that also bind many kinases at moderate affinity (Type 3 profiles in our taxonomy). For such compounds, ratio-based methods report high selectivity while distribution-based methods report moderate promiscuity, and both characterizations are correct given their respective questions.

This coexistence has a historical explanation. In classical pharmacology and analytical chemistry, selectivity is inherently pairwise. IUPAC formalizes it as a selectivity coefficient, the ratio of binding constants for an analyte and an interferent,[24] and ratio measures entered kinase profiling on that basis. Kinome-wide profiling, however, poses a distributional question: how concentrated is a compound's binding across hundreds of kinases? Entropy and Gini were introduced because ratio measures are ill-suited to it, since they discard all but two points of a 300–500 kinase profile. However, the older pairwise measures were never retired, leaving the field applying pairwise measures to a distributional question without a principled basis for choosing between them. The required properties proposed in the Required Properties section are an attempt to express what it means for a compound to be selective in a way that is precise enough to assess candidate selectivity definitions formally.

Practical Recommendations

Pending a formal resolution, three practical recommendations follow from our findings.

First, D1 (reliability threshold) should be adopted immediately as a reporting standard. Compounds with no detected binding above the assay detection floor should not receive selectivity scores, or should have their scores explicitly flagged as unreliable. Of the 222 Klaeger compounds, 16 (7.2%) fall into this category and currently receive scores that are indistinguishable from noise.

Second, selectivity assessments should report the sensitivity of rankings to parameter choice rather than a single score at a single parameter value. Our analysis shows that rank standard deviation across the parameter sweep is a compound-specific, interpretable quantity that identifies which compounds have robust selectivity assessments and which do not. This is analogous to reporting confidence intervals in statistical estimation.

Third, when comparing selectivity across studies that use different definitions, the comparison should be restricted to the subset of compounds and parameter values for which the definitions agree. The high within-family correlations ($r > 0.74$) suggest that S-score, entropy, and Gini are interchangeable for most purposes when parameters are chosen consistently.

However, ratio-based measures should not be compared directly with distribution-based measures without acknowledging the conceptual distinction.

Limitations

The analysis covers four profiling datasets spanning three assay technologies and two readout types, but these still represent a fraction of the chemical and kinase space relevant to drug discovery. The assay technologies are not directly comparable in absolute terms — the active-compound fractions differ across datasets for both genuine selectivity reasons and assay-specific detection thresholds. As a consequence, our analysis relies on within-dataset rank comparisons rather than cross-dataset absolute scores. The consistency of the two-family structure and the instability predictors across all four datasets indicates that the findings are not artifacts of a single assay platform. However, findings should still be interpreted in light of these technological differences.

We also examined whether in vitro selectivity predicts clinical toxicity, correlating the selectivity rankings of the FDA-approved Klaeger compounds with FAERS adverse-event rates ($n = 46$) and FDA-label discontinuation rates ($n = 33$). No association was significant under any definition (all $|r| < 0.27$, all $p > 0.13$). We do not read this as evidence that selectivity and toxicity are unrelated. FAERS counts mostly track how often a drug is prescribed, and how often a drug is withdrawn depends on the patient population treated for each indication, so neither measure can isolate the effect of selectivity. A proper test would require indication-stratified, treatment-related adverse-event rates from controlled trials with matched comparators, which are not available in standardized form across these drugs. Full details are given in the Supporting Information.

The four properties proposed above are necessary conditions motivated by empirical failure modes. The candidate measure shows that they are jointly satisfiable, but they are not proven to be sufficient to uniquely characterize a selectivity measure, and we do not show that they single out a measure of the candidate's form. The satisfaction assessments in Table 3 are based on analysis of the functional forms of each definition together with the empirical checks of this study. A formal proof of satisfaction or violation for each property–definition pair is left for future work.

Conclusion

We have shown that commonly used kinase inhibitor selectivity definitions cluster into two families that measure fundamentally different properties of binding profiles. We also showed that rank instability within and between families is mechanistically predictable from binding profile shape, and that three structurally distinct sources of instability can be identified and characterized. These findings motivate four required properties that any well-formed selectivity measure should satisfy: a reliability threshold condition, bounded gap sensitivity, distributional consistency, and monotonicity under weak off-target addition. No existing definition satisfies all four simultaneously, but we show that a gated, smooth-baseline effective-target-number does, and that it recovers its full-panel ranking from roughly half a typical panel.

The immediate practical implication is that selectivity assessments should report the sensitivity of rankings to parameter choice, and should explicitly flag compounds with no binding above the detection threshold as unreliable. The longer-term implication is that the field needs a principled basis for choosing among selectivity definitions rather than applying them interchangeably. Establishing whether these four properties characterize a selectivity measure uniquely, as Shannon's axioms do for entropy, is the natural next step.

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Supporting Information Available

Figure S1: operationalization-sensitivity check comparing the entropy and Gini rankings computed in forms close to the original publications (association-constant weighting) with the baseline-subtracted log-affinity forms used in the main text. Clinical outcome analysis: methods and full results of the secondary analysis relating selectivity rankings to FAERS adverse-event report rates and FDA-label discontinuation rates for the FDA-approved Klaeger compounds, including the per-definition correlations and the openFDA queries used. A preprint of this manuscript is available on ChemRxiv at `https://doi.org/10.26434/chemrxiv.15001618/v1`.

Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`; `https://doi.org/10.1038/nchembio.530`). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at `https://github.com/polinavino/kinase-selectivity-definitions`. No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code.

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Conflict of Interest

The author declares no conflict of interest.

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